

Strong operator vanishing of the staircase chain on $V^{(2,2,1^m)}$ (the $|S| = 0$ case of strong per-subset vanishing)

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Abstract

For $\lambda = (2, 2, 1^m)$ with $m \geq 2$, the staircase chain operator $\Omega := \prod_{k=1}^N (T_{i_k} + 1) \in H_q(S_n)$ (with $N = \binom{n}{2}$, $n = m + 4$, and (i_k) the staircase word $1; 2, 1; 3, 2, 1; \dots; n - 1, \dots, 1$) satisfies $\Omega = 0$ on V^λ as a linear operator. This is a strict strengthening of trace-vanishing $\text{tr}_{V^\lambda}(\Omega) = 0$ and follows in two lines from the Pillar 1 meta-theorem [1]: the column-1 chain image $B^{(\lambda)} \subseteq E_1^-$ is killed by the rightmost S_1 factor of the prefix block R'_ℓ . Equivalently, the $|S| = 0$ case of the strong per-subset operator vanishing conjecture holds.

For $|S| \geq 1$, the per-subset operator $M(S) := \prod_k A_k(S)$ (with $A_k(S) = (T_{i_k} + 1)/(q + 1)$ if $k \notin S$, $(qI - T_{i_k})/(q + 1)$ if $k \in S$) is conjecturally zero on V^λ for every S with $|S| < D := \tau(\tau + 1)/2$, $\tau = m - 1$. We verify the conjecture computationally for $m = 2, 3$ (a total of 233 subsets, all giving the zero matrix). Of the 232 non-trivial subsets at $m = 3$, exactly 229 follow the same Pillar 1 + S_1 -kill template that proves the $|S| = 0$ case (with a per-subset extension $M_T(S_T)V^\lambda \subseteq E_1^-$ that we verify empirically). The remaining 3 outliers escape this template — they have $M_T(S_T)V^\lambda$ in neither E_1^- nor E_2^- — but still vanish via inductive kill on $\tau \geq 1$ branching summands of $V^\lambda \downarrow S_\ell$. A clean characterisation covering all cases uniformly is left as a target.

1 Setup

We adopt the conventions of [1, 2, 3].

$H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$, quadratic relation $T_i^2 = (q - 1)T_i + q$, braid relations $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$, and commutation $T_i T_j = T_j T_i$ for $|i - j| \geq 2$. Write $S_i := T_i + 1$; the quadratic relation is equivalent to $S_i^2 = (q + 1)S_i$, so S_i has eigenvalues 0 and $q + 1$, and $\mathfrak{S}(S_i) = E_i^+ = \ker(T_i - q)$, $\ker(S_i) = E_i^- = \ker(T_i + 1)$.

Specht module and projectors. V^λ is the Specht module in the Hoefsmit ($b' = 1$) seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$. On V^λ , S_i is diagonalisable with eigenvalues $\{0, q + 1\}$, so $V^\lambda = E_i^+|_{V^\lambda} \oplus E_i^-|_{V^\lambda}$. The Hoefsmit local action of S_i on v_T depends only on the relative positions of $i, i + 1$ in T : SR gives $S_i v_T = (q + 1)v_T$, SC gives $S_i v_T = 0$, D gives a non-degenerate 2-block.

Projectors. Set

$$P_q^{(i)} := \frac{T_i + 1}{q + 1} = \frac{S_i}{q + 1}, \quad P_{-1}^{(i)} := \frac{qI - T_i}{q + 1} = I - P_q^{(i)}.$$

$P_q^{(i)}$ is the projector onto E_i^+ along E_i^- ; $P_{-1}^{(i)}$ is the projector onto E_i^- along E_i^+ .

Staircase word and chain. For $k \geq 1$, $R'_k := S_{k-1}S_{k-2} \cdots S_1$, with $R'_1 = 1$. The *staircase chain* for S_n is

$$\Omega := R'_2 R'_3 \cdots R'_n,$$

the product of S -factors along the reduced word $1; 2, 1; 3, 2, 1; \dots; n-1, \dots, 1$ for the longest element $w_0 \in S_n$. The total length is $N = \binom{n}{2}$.

Per-subset chain. For a subset $S \subseteq \{0, 1, \dots, N-1\}$ of word positions, define

$$M(S) := \prod_{k=0}^{N-1} A_k(S), \quad A_k(S) = \begin{cases} P_{-1}^{(i_k)} & k \in S, \\ P_q^{(i_k)} & k \notin S, \end{cases}$$

where i_k is the k th letter of the staircase word. Then $M(\emptyset) = \Omega/(q+1)^N$. The collection $\{M(S)\}$ is a “resolution of the identity” along the staircase: $\sum_S M(S) = \prod_k I = I$.

Target family. We focus on $\lambda = (2, 2, 1^m)$ with $m \geq 2$. Then $\widehat{\lambda} = (2, 2)$ ($\ell(\widehat{\lambda}) = 2$, $|\widehat{\lambda}| = 4$), $n = m + 4$, $\ell = \ell(\lambda) = m + 2$, and

$$\tau(\lambda) = \max(0, 2\ell(\widehat{\lambda}) + q - 1 - |\widehat{\lambda}|) = m - 1, \quad D(\lambda) := \frac{\tau(\tau + 1)}{2} = \frac{m(m - 1)}{2}.$$

Pillar 1’s standing hypothesis $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2 = 2$ becomes $m \geq 2$.

2 Main result and conjecture

Theorem 1 (Operator vanishing of Ω on V^λ). *For $\lambda = (2, 2, 1^m)$ with $m \geq 2$, the staircase chain operator*

$$\Omega = R'_2 R'_3 \cdots R'_n \in H_q(S_n)$$

acts as the zero operator on V^λ :

$$\Omega|_{V^\lambda} = 0.$$

Equivalently, the per-subset operator $M(\emptyset) = \Omega/(q+1)^N$ vanishes on V^λ , which is the $|S| = 0$ case of strong per-subset operator vanishing.

Conjecture 2 (Strong per-subset operator vanishing). *For $\lambda = (2, 2, 1^m)$ with $m \geq 2$ and any subset S with $|S| < D(\lambda) = m(m-1)/2$,*

$$M(S)|_{V^\lambda} = 0.$$

Theorem 1 is the $|S| = 0$ instance of Conjecture 2 (proved unconditionally below). Conjecture 2 is verified computationally on $m \in \{2, 3\}$: the empty subset for $(2, 2, 1, 1)$, and all $1+21+210 = 232$ subsets of size ≤ 2 for $(2, 2, 1, 1, 1)$, all giving the zero matrix [5].

3 Proof of Theorem 1

The proof rests on Pillar 1.

Theorem 3 (Pillar 1 meta-theorem [1]). *Let $\lambda = (\widehat{\lambda}, 1^q)$ with $\widehat{\lambda}$ having $\ell(\widehat{\lambda}) \geq 2$ rows, every row $\widehat{\lambda}_r \geq 2$, and $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. Set $\ell = \ell(\widehat{\lambda}) + q$, $\tau = q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$. Define the column-1 chain image*

$$B^{(\lambda)} := \Omega^{(\lambda)} \cdot V^\lambda, \quad \Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n.$$

Then

$$B^{(\lambda)} \subseteq \bigcap_{j=1}^{\tau} E_j^-|_{V^\lambda}.$$

In particular, $B^{(\lambda)} \subseteq E_1^-|_{V^\lambda}$ whenever $\tau \geq 1$.

For $\lambda = (2, 2, 1^m)$, $\widehat{\lambda} = (2, 2)$ satisfies the hypotheses (2 rows, each of length 2), and the standing hypothesis $q \geq 2$ is $m \geq 2$. So Theorem 3 applies.

Lemma 4 (S_1 -kill at the prefix–tail boundary). *R'_ℓ kills $E_1^-|_{V^\lambda}$: for any $v \in E_1^-|_{V^\lambda}$, $R'_\ell v = 0$.*

Proof. $R'_\ell = S_{\ell-1} S_{\ell-2} \cdots S_1$. Reading the product right-to-left in operator application, the rightmost factor is S_1 , so $R'_\ell v = S_{\ell-1} \cdots S_2(S_1 v) = S_{\ell-1} \cdots S_2(0) = 0$ by $E_1^- = \ker(S_1)$. $\square$ $\square$

Proof of Theorem 1. Split the staircase chain at the Pillar 1 boundary:

$$\Omega = (R'_2 R'_3 \cdots R'_\ell) \cdot \Omega^{(\lambda)} = R \cdot \Omega^{(\lambda)}, \quad R := R'_2 \cdots R'_\ell.$$

For $v \in V^\lambda$, $\Omega v = R(\Omega^{(\lambda)} v)$. By Theorem 3, $\Omega^{(\lambda)} v \in B^{(\lambda)} \subseteq E_1^-$. By Lemma 4, the rightmost factor of R , namely R'_ℓ , kills E_1^- . Therefore

$$R(\Omega^{(\lambda)} v) = R'_2 \cdots R'_{\ell-1} (R'_\ell(\Omega^{(\lambda)} v)) = R'_2 \cdots R'_{\ell-1} (0) = 0.$$

Hence $\Omega v = 0$ for every $v \in V^\lambda$, so $\Omega|_{V^\lambda} = 0$. $\square$ $\square$

Corollary 5 (Stronger half of trace-vanishing). *For $\lambda = (2, 2, 1^m)$, $m \geq 2$: $\text{tr}_{V^\lambda}(\Omega) = 0$ (because $\Omega|_{V^\lambda} = 0$, a fortiori). The trace-vanishing forward direction of [4, two-sided-sign-kill] is upgraded to a full operator-vanishing.*

4 Towards Conjecture 2: a per-subset Pillar 1

We now sketch the structure of the per-subset extension, identify the mechanism for 229/232 of the cases at $m = 3$, and document the obstruction in the remaining 3.

4.1 Per-subset prefix–tail split

For a subset $S \subseteq \{0, \dots, N-1\}$, split at position $\binom{\ell}{2}$:

$$M(S) = M_R(S_R) \cdot M_T(S_T),$$

where $S_R = S \cap \{0, \dots, \binom{\ell}{2} - 1\}$ and $S_T = S \cap \{\binom{\ell}{2}, \dots, N-1\}$. The prefix $M_R(S_R)$ and tail $M_T(S_T)$ are the per-subset analogues of R and $\Omega^{(\lambda)}$ respectively.

For $S = \emptyset$, $M_R = R/(q+1)^{\binom{\ell}{2}}$ and $M_T = \Omega^{(\lambda)}/(q+1)^{N-\binom{\ell}{2}}$, and the proof of Theorem 1 carries through. The general case faces two new issues:

- (i) **Per-subset Pillar 1:** does $M_T(S_T) \cdot V^\lambda \subseteq E_1^-$ still hold?
- (ii) **Prefix kill:** does $M_R(S_R)$ have a factor that kills E_1^- ? The rightmost factor of $M_R(S_R)$ is $A_{\binom{\ell}{2}-1}$ at letter s_1 . If $\binom{\ell}{2} - 1 \notin S$, this factor is $P_q^{(1)} = S_1/(q+1)$, which kills E_1^- . More generally, any position in S_R^c at letter s_1 in the prefix gives a kill chance.

4.2 Computational picture at $m = 3$

For $\lambda = (2, 2, 1, 1, 1)$, $\tau = 2$, $D = 3$, $\binom{\ell}{2} = 10$. The 232 subsets with $|S| < 3$ break into four bins (verified at $q = 5/7$ with exact rationals):

Table 1: Per-subset analysis at $\lambda = (2, 2, 1, 1, 1)$. “ $M_T \subseteq E_1^-$ ” means $\Im(M_T(S_T)) \subseteq E_1^-|_{V^\lambda}$, equivalently $S_1 \cdot M_T(S_T) = 0$. “Prefix has P_q at s_1 position” means at least one of the positions $\{0, 2, 5, 9\}$ (the in-prefix s_1 letters at the bottom of each prefix block) is *not* in S .

case	count	all have $M(S) = 0$?
$M_T \subseteq E_1^-$ AND prefix has $P_q^{(1)}$ at last-of-block	229	yes
$M_T \subseteq E_1^-$ AND prefix has no $P_q^{(1)}$ at last-of-block	0	—
$M_T \not\subseteq E_1^-$ AND prefix has $P_q^{(1)}$ at last-of-block	3	yes
$M_T \not\subseteq E_1^-$ AND prefix has no $P_q^{(1)}$ at last-of-block	0	—
Total	232	yes (verified)

The 229 “Pillar 1 template” subsets. For these, $M_T(S_T) \subseteq E_1^-$ AND at least one of the prefix’s “ s_1 end-of-block” positions (namely $\{0, 2, 5, 9\}$ for $\ell = 5$) is not in S . Among these, the sub-case where the *rightmost* prefix position $\binom{\ell}{2} - 1 = 9$ is not in S has a clean argument: there $A_9 = P_q^{(1)} = S_1/(q+1)$ is the first prefix factor applied (after the tail), and $A_9 \cdot E_1^- = 0$, so the running result vanishes immediately. This matches the proof of Theorem 1 verbatim.

The sub-case where $9 \in S$ but some earlier s_1 position $j^* \in \{0, 2, 5\}$ is in S^c is *not* a verbatim extension. After the tail’s E_1^- -valued output is multiplied on the left by $A_9 = P_{-1}^{(1)}$ (the identity on E_1^-), the next prefix factors at letters $s_2, s_3, \dots$ do not commute with T_1 in general and may move the running result out of E_1^- before reaching the $P_q^{(1)}$ -killing factor at j^* . So the “229 template” covers a clean sub-family by the $|S| = 0$ argument and the rest empirically; a genuine per-subset argument for the latter needs a refined containment beyond E_1^- (e.g. $\bigcap_{j=1}^{\tau-|S_T|} E_j^-$ from Conjecture 6, which is preserved across more prefix factors since each A_k at letter s_j has its image in $E_j^+ \cup E_j^-$).

The 3 outliers. The subsets $S = \{11, 12\}, \{12, 15\}, \{15, 16\}$ have $M_T(S_T) \not\subseteq E_1^-$. All three have $S \subseteq \{10, 11, \dots, 20\}$ (i.e., $S_R = \emptyset$), so $M_R(S_R) = R/(q+1)^{10}$ is the same R as in the $|S| = 0$ proof. For these, the kill mechanism is *induction on smaller shapes*:

- $V^\lambda \downarrow S_5 = V^{(2,2,1)} \oplus 2V^{(2,1,1,1)} \oplus V^{(1^5)}$.
- $R = \Omega_{S_5}/(q+1)^{10}$ is the staircase chain for S_5 acting on V^λ via the restriction.
- By induction (or direct application of Theorem 1 to smaller λ), $\Omega_{S_5} = 0$ on $V^{(2,1,1,1)}$ and $V^{(1^5)}$, the two $\tau \geq 1$ summands.
- Empirically, $M_T(S_T)$ has rank 1 for each of the three outliers, and $R \cdot M_T(S_T) = 0$ — the image of M_T projects into the kernel of $\Omega_{S_5}|_{V^{(2,2,1)}}$ (a 4-dimensional kernel in the 5-dimensional $\tau = 0$ summand).

Writing the $V^{(2,2,1)}$ -projection of $\Im(M_T)$ explicitly and verifying it lies in $\ker(\Omega_{S_5}|_{V^{(2,2,1)}})$ closes the $|S| = 2$ outliers, but the structural reason is not yet clean.

4.3 The gap to closing Conjecture 2

The clean statement we would like is:

Conjecture 6 (Per-subset Pillar 1, refined). *For $\lambda = (\hat{\lambda}, 1^q)$ in the Pillar 1 regime and any subset $S_T \subseteq \{\binom{\ell}{2}, \dots, N-1\}$ with $|S_T| < \tau$,*

$$M_T(S_T) \cdot V^\lambda \subseteq \bigcap_{j=1}^{\tau-|S_T|} E_j^-|_{V^\lambda}.$$

This would generalise Pillar 1 ($|S_T| = 0$ case) and explain the 229 template cases. The $|S_T| = \tau$ outliers escape this conjecture, but might fall into the inductive kill via Ω_{S_ℓ} .

Where Conjecture 6 would come from. Pillar 1’s proof reduces $B^{(\lambda)}$ recursively via the $\Phi_{k,*}$ -embedding to inner contributors $B^{(\mu_{k,*})}$, accumulated with the outer product $= \prod_{j=\ell}^{n-2} (T_j + 1)$. The threshold-matching identity (Pillar 1, Lemma 1) preserves τ across contributors. A per-subset version would track *which* of the $|S_T|$ “flips” occur in the outer product vs. inner contributor, and would show that each flip degrades the containment by at most one E_j^- level. Working this out is the key technical step.

5 Computational verification

We verify Conjecture 2 at $m = 2$ and $m = 3$ using exact rationals at $q = 5/7$ (and cross-checked at $q = 2/3$) with the Hoefsmit seminormal basis.

Table 2: Strong per-subset operator vanishing on $\lambda = (2, 2, 1^m)$ for $m \in \{2, 3\}$. Sweep over all S with $|S| < D$ confirms $M(S) = 0$ on V^λ as an exact rational matrix in every case.

λ	n	$\dim V^\lambda$	τ	D	# subsets $ S < D$	all zero?
$(2, 2, 1, 1)$	6	9	1	1	1	yes
$(2, 2, 1, 1, 1)$	7	14	2	3	232	yes

At $m = 2$, $D = 1$ and the only subset is $S = \emptyset$, covered by Theorem 1. At $m = 3$, the kill index distribution shows all 232 subsets short-circuit deeply: the running product $\text{Pref}_k := A_0 A_1 \cdots A_{k-1}$ hits the zero matrix at step $k^*(S)$, with kill index distribution

kill step	letter	factor type	count
11	s_5	$P_q^{(5)}$	172
16	s_6	$P_q^{(6)}$	57
17	s_5	$P_q^{(5)}$	3

The killing factor is always $P_q^{(j)}$ for $j \in \{\ell, \ell + 1\} = \{5, 6\}$, and the running rank just before the kill is always 1 (rank-1 line lands in E_j^-). This is a much more rigid structure than Pillar 1 directly states.

6 Status

Established (this paper).

- Theorem 1: $\Omega = 0$ on V^λ for $\lambda = (2, 2, 1^m)$, $m \geq 2$. (Equivalently, $|S| = 0$ case of Conjecture 2.) The proof is a two-line corollary of Pillar 1.
- Computational verification of Conjecture 2 for $m \in \{2, 3\}$ (233 subset operators total).
- Empirical characterisation of the kill structure at $m = 3$: all kills are by $P_q^{(j)}$ for $j \in \{\ell, \ell+1\}$, at the first appearance of T_j in the staircase (with one exception at step 17).
- 229/232 subsets at $m = 3$ fit the “Pillar 1 + S_1 -kill template” (per-subset extension $M_T(S_T) \subseteq E_1^-$).

Open (residual gaps).

- **Conjecture 6:** a per-subset Pillar 1 giving $M_T(S_T) \subseteq E_1^-$ for $|S_T| < \tau$. Would cover the 229 template cases uniformly across m .
- **The 3 outliers at $m = 3$:** subsets $\{11, 12\}, \{12, 15\}, \{15, 16\}$ have $M_T(S_T) \not\subseteq E_1^-$ but $M(S) = 0$ via the inductive kernel of Ω_{S_ℓ} on the $\tau = 0$ branching summand. A structural reason (rather than case-by-case verification) is needed.
- **Extension to $m \geq 4$:** computational sweep over $m = 4$ ($D = 6$, large search space) and inductive proof of the full conjecture.

Methodological note. This proof session settled the $|S| = 0$ case as a clean corollary of Pillar 1 once we recognised the right split (prefix R via $R'_2 \cdots R'_\ell$, tail $\Omega^{(\lambda)}$). The per-subset extension stalled at the 3 outliers at $m = 3$: the “Pillar 1 + S_1 -kill” template is sharp but not complete. The remaining case requires a per-subset version of Pillar 1 itself, which in turn requires extending the multi-row meta-theorem to mixed P_q/P_{-1} chains — a non-trivial undertaking deferred to a future session.

References

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